
ERNEST P. CHAN · MACHINE TRADING, WILEY 2016

Options Strategies

A teaching walkthrough of Chapter 5 — six volatility strategies, what they actually returned, and why execution decides all of them.

6 strategies · 2004–2016 test windows · 9 exercises

THE CENTRAL CLAIM

Consistent alpha comes from
selling volatility.

Long volatility is a structural profit drag —
carried only as tail-risk insurance, never as
the thesis.

How to read this deck

Every number on the following slides is taken from Chapter 5's own backtests. They are *published research results*, not live track records — the chapter is unusually candid that several of them do not survive contact with a real order book.

NOTATION USED THROUGHOUT

VX	CBOE VIX futures — front month unless stated
VXX / XIV	Long / inverse short-term VIX futures ETPs
CL / LO	WTI crude futures / American options on CL
RV, IV	Realized and implied volatility
Calmar	CAGR ÷ maximum drawdown — the chapter's preferred yardstick

READ EVERY RESULT THROUGH THESE THREE FILTERS

- 01 Was it filled at the mid?
Option spreads run 1–5% of underlying price. Mid-price backtests can invert a strategy's entire sign.
- 02 What is the denominator?
Returns on option market value flatter deep-OTM books enormously — embedded leverage, not skill.
- 03 Where does the tail live?
Short-vol drawdowns of 50–92% are normal here. Ask what caps the loss before you ask what it earns.

Six strategies, one premium

Each strategy harvests the same variance risk premium through a different instrument and a different risk container.

STRATEGY 01

Volatility without options

Short VX futures, XIV and SPY. Harvest contango roll yield, then tame the drawdown with a Kalman-filtered hedge ratio.

STRATEGY 02

GARCH volatility forecasting

Predict tomorrow's realized volatility at ~69% accuracy — then trade the opposite of the naive interpretation.

STRATEGY 03

Event-driven volatility

Weekly EIA petroleum reports. Buying the event loses badly; selling the calm window between events pays.

STRATEGY 04

Gamma scalping

Short volatility via a mean-reversion grid, wrapped inside a long strangle that hard-caps the tail at \$4,000.

STRATEGY 05

Dispersion trading

Short the index straddle, own 50 single-stock straddles. Delta- and vega-neutral: pure short implied correlation.

STRATEGY 06

Cross-sectional IV reversion

Rank the whole S&P 500 option universe daily. A cautionary tale about embedded leverage in the denominator.

Three rules that shape every trade in the chapter

I

Sell the insurance, don't buy it

The repeatable edge is the **variance risk premium**: options are structurally rich because someone must be paid to carry crash risk. Long option books pay negative theta every single day for the privilege of being right occasionally.

II

Never buy delta through an option

If you want directional exposure, buy the underlying — the spread is a fraction of the option's. Options earn their keep only on the dimensions the underlying cannot express: **volatility, theta, skew, implied correlation**.

III

Frictions decide, not signals

Option bid-ask spreads run **1% to 5%+ of the underlying price**. A backtest that fills at the mid shows spectacular returns that vanish the moment a live order crosses the spread twice.

*The corollary the rest of the chapter keeps proving: a strategy is not the signal — it is the signal **minus the spread** you must cross to express it.*

Equities are a time series. Options are a tensor.

THE SEARCH SPACE

Underlyings (N) × Expirations (T) × Strikes (K)
× Call/Put × Bid/Ask

A full S&P 500 chain across all strikes and expirations over ten years *will not fit in memory* as a dense array. Before you have a strategy, you have a data-engineering problem.

FOUR HURDLES THAT HAVE NO EQUITY ANALOGUE

01 Dimensionality & memory

Chains must be compressed daily into a handful of dense feature arrays — ATM index, total theta, total vega, position.

02 Expiration & roll logic

Every continuous position needs programmatic roll rules. Contracts die; the strategy must not.

03 Path dependency of the Greeks

Δ , Γ , Θ and vega all drift with spot and elapsed time, so exposure must be rebalanced rather than held.

04 Asymmetric liquidity

Tick data updates at microsecond scale and prints inside — or outside — a wide best bid/offer.

1. Trading Volatility Without Options (Pseudo–Options Trading)

STRATEGY 01

Trading volatility without options

Skip the option chain entirely. VIX futures and volatility ETPs give you the same premium in a single, screen-traded leg — and the term structure hands you the carry.

VIX futures sit in **contango roughly 80–85% of the time**. The front contract must converge down toward spot as expiry approaches, so a systematic short collects that decay as roll yield.

COVERED IN THIS SECTION

- Short VX vs. long SPY, sized by Kelly
- VX-ES roll-return regime timing
- Kalman-filtered dynamic hedge, XIV vs. SPY

FROM INDEX OPTIONS TO A TRADABLE SHORT

S&P 500 (SPX) option chain

| CBOE 30-day model



VIX index (spot)

NOT TRADABLE

| term structure



VX futures – front / back

Short VX / XIV

Captures contango roll yield

Long VXX

Suffers structural roll drag

Short VX beats long SPY — at the same brutal cost

Test window
2004-04-05 → 2015-08-19

POSITION SIZING — THE KELLY CRITERION

$$f^* = \frac{\mu}{\sigma^2} = \frac{\mathbb{E}[r]}{\text{Var}(r)}$$

Mean return divided by variance gives the growth-optimal leverage. Applied to each return series independently, it produces very different verdicts.

SPY OPTIMAL
LEVERAGE

+2.15×

VX SHORT OPTIMAL
LEVERAGE

−0.88×

% Sizing via Kelly criterion

```
kelly_vx = mean(ret_VX) / var(ret_VX); % -0.88
kelly_spy = mean(ret_SPY) / var(ret_SPY); % +2.15
```

KELLY-LEVERAGED PERFORMANCE, 2004–2015

METRIC	2.15× LONG SPY	−0.88× SHORT VX
CAGR	7.2%	17.8%
Max drawdown	−86.3%	−91.8%
Calmar ratio	0.084	0.19

Short volatility earns more than **2.4× the equity risk premium** and roughly double the Calmar — but a −91.8% drawdown is not an allocation, it is a liquidation. Both series break in the same crash.

So the next two refinements attack the drawdown, not the return.

Two refinements take Calmar from 0.19 to 0.97

STATIC SHORT VX

0.19

+ ROLL-RETURN TIMING

0.55

+ KALMAN DYNAMIC HEDGE

0.97

B · VX-ES ROLL-RETURN REGIME TIMING

$$\text{Roll return} = \frac{F_1 - F_2}{F_1}$$

$\leq -10\%$ deep backwardation, panic → **buy VX, long ES**
 $\geq +10\%$ steep contango, calm bull → **short VX, short ES**

Contract ratio 0.3906 : 1. The regime rule stops the strategy from being maximally short exactly when volatility is exploding.

Timing trap: VX settles 16:15 ET, ES Globex closes 16:00 ET. Signals must use lagged closes or the backtest looks ahead.

C · KALMAN-FILTERED HEDGE, XIV VS. SPY

A fixed 0.3906 hedge ratio is wrong almost always: high-volatility regimes overestimate it, low-volatility regimes underestimate it. Let the data re-estimate β every day.

$$\beta_t = \beta_{t-1} + w_t, \quad w_t \sim \mathcal{N}(0, W)$$

$$r_{\text{XIV},t} = \beta_t r_{\text{SPY},t} + v_t, \quad v_t \sim \mathcal{N}(0, V)$$

Recursive Bayesian updating — a hidden state that walks, observed through a noisy regression; it only keeps the hedge honest.

Calmar **0.97** over 2010-11-30 → 2015-08-19 — the chapter's best risk-adjusted volatility trade.

2. Volatility Forecasting with GARCH & The RV vs. IV Disconnect

STRATEGY 02 · A

GARCH forecasts volatility well. That is the easy half.

SPECIFICATION — GARCH(P, Q)

$$r_t = \sigma_t \varepsilon_t, \quad \varepsilon_t \sim \text{i.i.d. } \mathcal{N}(0, 1)$$

$$\sigma_t^2 = \omega + \sum_{i=1..p} \alpha_i \sigma_{t-i}^2 + \sum_{j=1..q} \beta_j r_{t-j}^2$$

Tomorrow's variance is a weighted blend of yesterday's **forecast** variance and yesterday's **squared surprise**. Volatility clusters, so this simple recursion carries real information.

MODEL SELECTION

A **10 × 9 grid** of (p, q) fitted on SPY returns, 2005–2011.

Lowest BIC selects **GARCH(1, 2)** — parsimony wins over the richer specifications.

DIRECTIONAL ACCURACY — SIGN OF $\Delta \Sigma_{T+1}$

TRAIN SET

72%

SPY OUT-OF-SAMPLE

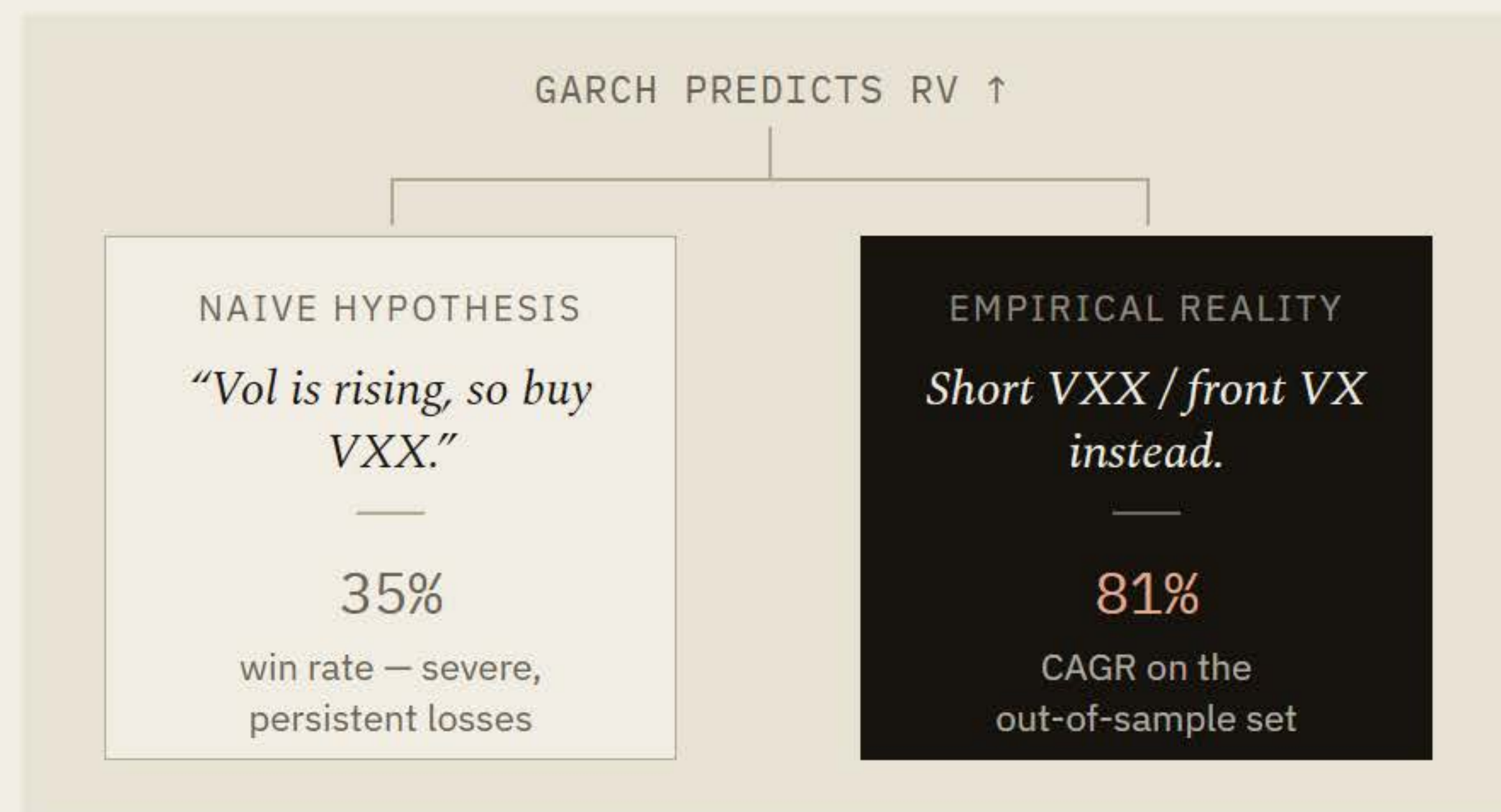
69%

OTHER UNDERLYINGS, 2011–2015 TEST SET

USO	67%
EURUSD	62%
AAPL	60%
GLD	59%

69% accuracy on realized volatility — and yet the obvious trade built on it loses money fast. Next slide.

Volatility rises. Your long volatility position still loses.



WHY REALIZED VOLATILITY FIGHTS IMPLIED VOLATILITY

1 · Theta decay is relentless

VXX tracks front and second-month VX futures. Even with market volatility elevated, roll and time decay erode the ETP every single day.

2 · Asymmetric spot-vol correlation

Equity rallies crush implied volatility even when intraday price dispersion — the thing GARCH actually forecasts — is large.

3 · The measured disconnect

Daily SPY return magnitude moves with VXX only **35% of the time** — just 43% on down days.

The lesson generalises past volatility: a correct forecast of the wrong variable is not an edge. Forecast the instrument you will actually hold.

Two ways to trade the same forecast

Same GARCH(1,2) engine.
Different benchmark for “high”.

MODEL A

Compare the forecast to yesterday

$$RV_{t+1} - RV_t$$

If GARCH says $\sigma_{t+1} > \sigma_t$, **short VXX** at the close. If $\sigma_{t+1} < \sigma_t$, **buy VXX**. Hold one day. It is the paradox from the previous slide, traded deliberately.

CAGR

81.0%

CALMAR RATIO

1.90

Out-of-sample 2011–2015. The strongest Calmar in the entire chapter — on a single-leg ETP trade with no option chain at all.

MODEL B · AHMAD, 2005

Compare the forecast to the market's price

$$RV_{t+1} - VIX_t$$

If forecast RV exceeds spot VIX, **buy the front VX future** — volatility is cheap relative to what is coming. Otherwise **short VX**. A relative-value read, not a momentum read.

CAGR

41.7%

CALMAR RATIO

0.50

2010–2016. **Decayed sharply after 2014** as institutional volatility-targeting flows compressed the very spread it traded.

Model B's decay is the more instructive result: crowding erodes a spread trade long before the model breaks.

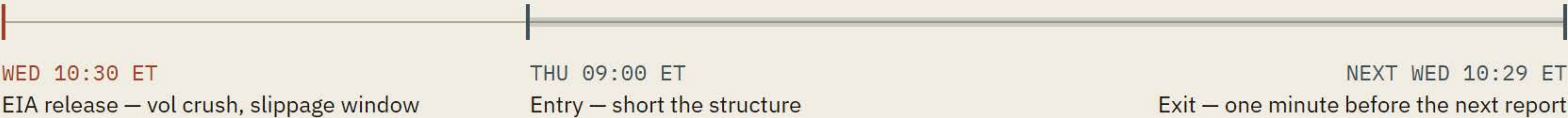
3. Event–Driven Strategies (e.g., EIA Petroleum Reports)

STRATEGY 03 · EVENT-DRIVEN

Don't trade the event.
Trade the quiet after it.

EIA Petroleum Status Report
CL futures · L0 options
Nanex tick data, 25 ms stamps

THE WEEKLY CYCLE



the calm window — six days of theta with no scheduled catalyst

TRADING THROUGH THE EVENT — BOTH DIRECTIONS LOSE

Long straddle into release

Wed 09:00 → Wed 10:31

-\$27,110

Short straddle into release

crosses the spread twice
Per straddle, per year. Buying pays for a marked-up IV that collapses; selling pays the spread twice with market orders.

-\$36,060

SELLING THE CALM WINDOW — ALL THREE VARIANTS PAY

Short ATM straddle

+\$13,270 DD \$4,050

Short 5% OTM strangle

+\$10,640 DD \$3,410

Exit before API, Tue 16:29

+\$9,750 DD \$2,950

Exiting earlier to dodge the Tuesday API report cuts both profit and drawdown — a cleaner risk-adjusted trade.

4. Gamma Scalping

STRATEGY 04

Short volatility inside a hard-capped box

CL futures grid

+ long LO strangle

Thu 09:00 → Fri 14:30 ET

Two positions that are each individually unattractive: a long strangle that bleeds theta, and a mean-reversion grid that can run away from you.

Combined, the strangle's gamma *pays for* the grid's tail.

Long 5% OTM strangle

$\Gamma > 0$ $\nu > 0$ $\theta < 0$

The insurance leg. Costs theta daily; converts to a linear hedge beyond $\pm 5\%$.

+

Intraday grid on CL

short at $+1\%$, $+2\%$, ... $+N\%$

long at -1% , -2% , ... $-N\%$

The engine. Sells each oscillation around the entry price.

WHY THE TWO LEGS NEUTRALISE EACH OTHER

CL rallies hard

Call delta $\rightarrow +1$, put delta $\rightarrow 0$. The short futures stacked at the upper grid rungs offset the call. Net delta stays near zero.

CL collapses

Put delta $\rightarrow -1$, call delta $\rightarrow 0$. The long futures accumulated at the lower rungs offset the put. Same result, mirrored.

ANNUAL P&L

ATM straddle + grid, $N=1$

negative

ATM theta outruns the scalps

5% OTM strangle + grid

+\$6,370

max DD \$9,400

loss capped at
\$4,000

STRATEGY 05

Selling the correlation everyone overpays for

WHY THE INDEX OPTION IS STRUCTURALLY RICH

$$\sigma_I^2 = \sum w_i^2 \sigma_i^2 + 2 \sum \sum w_i w_j \sigma_i \sigma_j \rho_{ij}$$

Index variance is component variance *plus* a correlation term. Institutions buy index puts for portfolio insurance, bidding up that correlation term — so **implied correlation sits persistently above realized.**

Sell the index straddle, buy the components: you are short implied correlation, nothing else.

CONSTRUCTION, IN ORDER

- 01 Screen the S&P 500 daily for tenor ≥ 28 days and $|\Delta| \leq 0.25$
- 02 Keep the **50 names with the highest (least negative) theta** — minimise the decay you pay
- 03 Size each leg to control \$1 of underlying, then short SPX to vega-

neutral 2007–2013, on gross market value. **Barely dented by Lehman 2008 or the August 2011 downgrade** — component gains offset the index short exactly when correlation spiked.

VEGA-NEUTRAL SIZING

$$\text{Units}_i = \frac{1}{S_i \times 100}$$

$$v_{\text{stocks}} = \sum \text{Units}_i \times v_i$$

$$\text{Pos}_{\text{SPX}} = - \frac{v_{\text{stocks}}}{v_{\text{SPX}}}$$

ARRAY COMPRESSION — 500 CHAINS INTO FOUR DENSE ARRAYS

T×M raw chains → ATMidx(t, c, call/put)
strikes, expiries, totTheta(t, c)
Greeks, quotes totVega(t, c)
pos(t, c)

CAGR	MAX DD	CALMAR
19.0%	51%	0.40

6. Cross-Sectional Mean Reversion of Implied Volatility

STRATEGY 06 · A CAUTIONARY TALE

1.1% a day — and almost none of it is real

THE RULE

Every day, across the S&P 500 option universe with matching tenors **≥ 28 days**: buy the **50 calls with the lowest IV**, short the **50 puts with the highest IV**. Hold one day. No directional forecast — pure relative value.

REPORTED DAILY RETURN

1.1%

ANNUALISED, 2007–2013

>250%

THE MECHANISM THAT MANUFACTURES THE NUMBER

$$\Omega = |\Delta| \times \frac{S_{\text{underlying}}}{P_{\text{option}}}$$

As an option moves deeper OTM, its price collapses toward zero **faster than its delta does**. Embedded leverage explodes. Dividing P&L by that vanishing market value inflates the percentage return without adding a dollar of profit.

THREE REASONS THE BACKTEST IS AN ILLUSION

1 · A denominator, not a return

Returns computed on option market value measure embedded leverage, not skill. The same dollars over a smaller base look like genius.

2 · Deep-OTM spreads are enormous

In percentage terms the bid-ask on these contracts dwarfs the edge. Mid-price fills are not an approximation here — they are fiction.

3 · Unhedged delta and vega

The book ends up net long delta and net short vega. Much of the “alpha” is uncompensated factor exposure waiting for a bad week. *Chan includes this strategy precisely because it looks best on paper. It is the chapter's warning about its own methodology.*

The full scoreboard

Test windows differ by strategy — compare Calmar, not CAGR. Dashes are figures the chapter does not report.

STRATEGY	INSTRUMENTS	WINDOW	ALPHA SOURCE	CAGR	MAX DD	CALMAR
Short VX (Kelly)	VX front-month futures	2004–15	Contango roll yield	17.8%	−91.8%	0.19
VX–ES roll timing	VX + ES futures	2004–15	Term-structure slope	—	—	0.55
Dynamic XIV / SPY	XIV ETN + SPY	2010–15	Kalman dynamic hedge	—	—	0.97
GARCH $RV_{t+1} - RV_t$	VXX short / long	2011–15	RV vs. IV disconnect	81.0%	—	1.90
GARCH $RV_{t+1} - VIX_t$	VX futures	2010–16	Spread mispricing	41.7%	—	0.50
Post-EIA strangle	LO options on CL	1 yr	Calm-window theta	+\$10.6k	−\$3.4k	~3.1
Gamma scalping	CL futures + strangle	1 yr	Grid mean reversion	+\$6.4k	−\$9.4k	~0.68
SPX dispersion	SPX + 50 straddles	2007–13	Implied correlation	19.0%	−51.0%	0.40
Cross-sectional IV	50 calls / 50 puts	2007–13	IV reversion (paper)	1.1%/day	—	n/a

Read down the Calmar column, not the CAGR column: the two highest scores both come from containing the drawdown — a dynamic hedge and a one-day holding period — not from a better forecast.

Five ways a good strategy dies in production

None of these are modelling errors.
All of them are engineering errors.

01 Mid-price bias

Never evaluate an options backtest on mid quotes. Model explicit slippage, or simulate passive limit-order fill probabilities. This single assumption flips signs.

02 Timestamp desynchronisation

VX closes 16:15 ET; ES closes 16:00 ET. Any cross-market signal must use strictly lagged prices or the backtest is quietly looking ahead.

03 Corporate action traps

Splits change the contract's deliverable multiplier. Long calls receive no dividend compensation — price simply drops on the ex-date.

04 Delta drift in gamma scalping

A static straddle does not stay delta-neutral as spot moves. Grid rebalancing must track the option's shifting delta, not a fixed schedule.

05 Short-volatility tail risk

Unhedged short-vol ETPs face **instant termination**, not merely drawdown — XIV was liquidated in February 2018, after this book was published.

THE THROUGH-LINE

*Profitable live deployment needs an intelligent limit-order routine that earns fills near the mid. The execution layer **is** the strategy.*

Worked answers, in one page

Two are analytic (5.3, 5.9); the rest are backtest extensions whose *direction* follows from the chapter's own logic.

5.1 · SIGN OF VEGA

Strictly positive — $\mathcal{V} = S\sqrt{T} \cdot \phi(d_1) > 0$

Vega measures sensitivity to *implied* volatility. The anti-correlation is with *realized* volatility, a different variable — plus roll and theta decay erode value regardless of vega.

5.2 · EVENT-RULE REFINEMENTS

Filter on premium richness

Sell only when $IV_{30} / HV_{30} > 1.25$; maximise $|\Theta|$ / margin; take profit at +50% of credit, hard stop at –100% (2× credit) to truncate gap risk.

5.3 · MAX LOSS PROOF, $N = 1$

Exactly \$4,000 per contract

Buy at –1% (\$99), liquidate at the 5% boundary (\$95) → $\$4.00/\text{bbl} \times 1,000 \text{ bbl} = \$4,000$. Mirrored on the upside: short \$101, cover \$105. Beyond ±5% the long option hedges dollar-for-dollar.

5.4 · DELTA-HEDGE WITH SPY

Short $\Delta_{\text{port}} / S_{\text{SPY}}$ shares daily

Portfolio delta is net positive (long calls – short puts). Hedging removes beta drift and truncates 2008-style drawdowns, at a moderate cost to gross return in strong bull regimes.

5.5 · LONG-ONLY VERSION

Negative long-run Sharpe

Buying only the lowest-IV calls leaves the book heavily long delta and long vega, with no short premium to offset daily theta. Decay dominates.

5.6 · VEGA-NEUTRAL WEIGHTING

$N_{\text{SPX}} = -\sum \mathcal{V}_{\text{call},i} / \mathcal{V}_{\text{SPX}}$

Shorting index calls against long stock calls turns Exercise 5.5 into a single-leg dispersion trade: macro IV risk removed, single-name relative value retained.

5.7 · HOURLY BARS

More positive intraday

RV and VXX correlate more positively during market hours, especially in morning shocks — but overnight gaps and afternoon theta bleed keep the daily total negative.

5.8 · VXX INSTEAD OF VX

Yes, with an asymmetric threshold

Long VXX if forecast RV > VIX, else short. Because roll decay is a tailwind for the short, the long signal needs a buffer: $RV - VIX > c$.

5.9 · SECOND-ORDER DELTA

Delta falls by $\Gamma \cdot \Delta S$

Taylor expansion: $\delta(\Delta) \approx (\partial \Delta / \partial S) \cdot \Delta S = \Gamma \cdot \Delta S$. For a drop of $-\Delta S$ the change is $-\Gamma \cdot \Delta S$.

Four things to carry out of this chapter

01

Be the insurer, and price the ruin

Selling volatility is where the premium lives. But a -91.8% drawdown means the naive version is uninvestable. Every good result in the chapter comes from capping the tail, not from predicting better.

02

Forecast the thing you will hold

69% accuracy on realized volatility loses money when the instrument is VXX. Realized and implied volatility are different variables — and only one of them settles your P&L.

03

Sell the calm, never the catalyst

Trading through the EIA release loses \$27k–\$36k a year in either direction. Holding the six quiet days between releases earns \$10k–\$13k. Same premium, opposite window.

04

Distrust your best backtest most

The 250%-a-year result is the one that is fake. Interrogate the denominator, the fill price and the unhedged Greeks before you celebrate.

The chapter's real subject is not volatility. It is the gap between a backtest and a fill — and the engineering required to close it.

Next in the book: intelligent limit-order execution — the machinery that decides whether any of the above survives contact with the market.